1 Important Inequalities

The proofs of the error bound formulas all use the *Triangle Inequality*

$$\left|\sum_{i=1}^{n} x_i\right| \leq \sum_{i=1}^{n} |x_i|,$$

which you proved in AEP #1, as well as the following similar-looking inequality for integrals.

Proposition 1.1 (Triangle inequality for integrals)**.** *If $f : [a,b] \to \mathbb{R}$ is integrable then so is $|f|$ and*

$$\left|\int_a^b f(x)dx\right| \leq \int_a^b |f(x)|dx.$$

1. Assume the first statement in Proposition 1.1 and prove the second.
(*Hint*: $-|f(x)| \leq f(x) \leq |f(x)|$.)

2 The Left- and Right-endpoint Rules

Theorem 2.1 (Error bound for left- and right-endpoint approximations)**.** *Let $E_n^L(f)$ and $E_n^R(f)$ denote the absolute error $\left|\sum_{i=1}^n f(x_i^*)\Delta x - \int_a^b f(x)dx\right|$ using the left- and right-endpoint approximations, respectively. If $M = \max\{|f'(x)|\}_{x\in[a,b]}$, then E_n^L and E_n^R are both at most $M\frac{(b-a)^2}{2n}$.*

Proof. (a) For the left-endpoint approximation, show that the absolute error $E_n^L(f)$ can be written as

$$E_n^L(f) = \left|\sum_{i=1}^n L_i\right|,$$

where

$$L_i := f(x_{i-1})\left(\frac{b-a}{n}\right) - \int_{x_{i-1}}^{x_i} f(x)dx. \tag{2.2}$$

(*Hint*: Start with the definition $E_n^L(f) = \left|\sum_{i=1}^n f(x_i^*)\Delta x - \int_a^b f(x)dx\right|$ and write $\int_a^b f(x)dx = \sum_{i=1}^n \int_{x_{i-1}}^{x_i} f(x)dx$.)

(b) Show that Equation (2.2) can be written as

$$L_i = \int_{x_{i-1}}^{x_i} (t - x_i) f'(t) dt. \tag{2.3}$$

(*Hint*: Integrate Equation (2.3) by parts.)

(c) Assume now that $M = \max\{|f'(x)|\}_{x\in[a,b]}$. Substitute Equation (2.3) into $E_n^L(f) = |\sum_{i=1}^n L_i|$ and use the Triangle Inequality together with Proposition 1.1 to show that

$$E_n^L(f) \leq M\frac{(b-a)^2}{2n}.$$

The proof that $E_n^R \leq M\frac{(b-a)^2}{2n}$ is essentially the same, since in this case $L_i = \int_{x_{i-1}}^{x_i}(t-x_{i-1})f'(t)dt$. □

3 The Midpoint Rule

Theorem 3.1 (Error bound for the midpoint approximations)**.** *Let $E_n^M(f)$ denote the absolute error $|\sum_{i=1}^n f(x_i^*)\Delta x - \int_a^b f(x)dx|$ for the the midpoint approximation. If $M = \max\{|f''(x)|\}_{x\in[a,b]}$, then $E_n^M(f) \le M\frac{(b-a)^3}{24n^2}$.*

To prove this theorem, we will need the following lemma.

Lemma 3.2.

$$f(b) - f(a) = \int_a^b f'(x)dx = \int_0^1 f'(a+t(b-a))(b-a)dt.$$

(a) Prove Lemma 3.2. (*Hint*: Use substitution.)

We now prove Theorem 3.1.

Proof. (b) For the midpoint approximation, show that the error can be written as

$$E_n^M(f) = \left|\sum_{i=1}^n L_i\right|,$$

where

$$L_i := \int_{x_{i-1}}^{x_i} f(x)dx - f\left(\frac{x_{i-1}+x_i}{2}\right)(x_i - x_{i-1}). \tag{3.3}$$

(c) Use Lemma 3.2 to show that Equation (3.3) can be written as

$$L_i = \int_{x_{i-1}}^{x_i} \left[\int_0^1 f' \left(\frac{x_{i-1}+x_i}{2} + t\left(x - \frac{x_{i-1}+x_i}{2} \right) \right) dt \right] \left(x - \frac{x_{i-1}+x_i}{2} \right) dx. \qquad (3.4)$$

(*Hint*: $\left(x - \frac{x_{i-1}+x_i}{2} \right)$ is independent of the integration variable t.)

(d) Integrate Equation (3.4) by parts by taking

$$u = \int_0^1 f'\left(\frac{x_{i-1}+x_i}{2} + t\left(x - \frac{x_{i-1}+x_i}{2}\right)\right) dt,$$
$$dv = \left(x - \frac{x_{i-1}+x_i}{2}\right) dt,$$

to show that

$$L_i = -\int_{x_{i-1}}^{x_i} \frac{d}{dx}\left[\int_0^1 f'\left(\frac{x_{i-1}+x_i}{2} + t\left(x - \frac{x_{i-1}+x_i}{2}\right)\right) dt\right] \frac{1}{2}(x-x_{i-1})(x-x_i). \qquad (3.5)$$

(*Hint:* Note that we can take v to be any antiderivative of $\left(x - \frac{x_{i-1}+x_i}{2}\right)$. Things work out most nicely by taking $v = \frac{1}{2}(x-x_{i-1})(x-x_i)$.)

(e) Assuming that you can move the derivative under the integral sign (after taking MVC you will be able prove this step is valid); i.e.,

$$\frac{d}{dx}\int_0^1 f'\left(\frac{x_{i-1}+x_i}{2}+t\left(x-\frac{x_{i-1}+x_i}{2}\right)\right)dt=\int_0^1 \frac{d}{dx}f'\left(\frac{x_{i-1}+x_i}{2}+t\left(x-\frac{x_{i-1}+x_i}{2}\right)\right)dt,$$

show that Equation (3.5) becomes

$$L_i=-\int_{x_{i-1}}^{x_i}\left[\int_0^1 f''\left(\frac{x_{i-1}+x_i}{2}+t\left(x-\frac{x_{i-1}+x_i}{2}\right)\right)tdt\right]\frac{1}{2}(x-x_{i-1})(x-x_i)dx. \qquad (3.6)$$

(f) Now assume $M = \max\{|f''(x)|\}_{x\in[a,b]}$ and substitute Equation (3.6) into $E_n^M(f) = |\sum_{i=1}^n L_i| \leq \sum_{i=1}^n |L_i|$ to derive the bound

$$E_n^M(f) \leq M\frac{(b-a)^3}{24n^2}.$$

□

4 The Trapezoidal Rule

Theorem 4.1 (Error bound for the trapezoidal approximation)**.** *Let $E_n^T(f)$ denote the absolute error $|\sum_{i=1}^n f(x_i^*)\Delta x - \int_a^b f(x)dx|$ using the trapezoidal approximation. If $M = \max\{|f''(x)|\}_{x\in[a,b]}$, then $E_n^T \leq \frac{K(b-a)^3}{12n^2}$.*

Proof. (a) Show that the error can be written as

$$E_n^T(f) = \left|\sum_{i=1}^n L_i\right|,$$

where

$$L_i := \left(\frac{f(x_{i-1})+f(x_i)}{2}\right)\frac{b-a}{n} - \int_{x_{i-1}}^{x_i} f(x)dx. \tag{4.2}$$

(b) Show that Equation (4.2) can be rewritten as

$$L_i = \int_{x_{i-1}}^{x_i} \left(t - \frac{x_{i-1} + x_i}{2} \right) f'(t)dt. \tag{4.3}$$

(*Hint*: Integrate Equation (4.3) by parts.)

(c) Integrate Equation (4.3) another way to show that

$$L_i = \frac{1}{2}\int_{x_{i-1}}^{x_i}\left(\left(\frac{b-a}{2n}\right)^2 - \left(t - \frac{x_{i-1}+x_i}{2}\right)^2\right) f''(t)dt.$$

(d) Now assume that $|f''(t)| \leq M$ for all $t \in [a,b]$ and finish the derivation.

□

5 Simpson's Rule

Theorem 5.1 (Error bound for Simpson's Rule)**.** *Let $f(x)$ be a continuous function over $[a,b]$ having a fourth derivative $f^{(4)}(x)$ over this interval. If $M = \max\{|f^{(4)}(x)|\}_{x\in[a,b]}$, then the upper bound for the error in approximating $\int_a^b f(x)dx$ using Simpson's Rule is*

$$E_n^S(f) \leq \frac{M(b-a)^5}{180n^4}.$$

Proof. (a) Consider the piecewise function [1]

$$p(x) = \begin{cases} (x-x_{2i-2})^3(x-\frac{x_{2i-2}+2x_{2i}}{3}), & x_{2i-2} \leq x \leq x_{2i-1}, \\ (x-x_{2i})^3(x-\frac{2x_{2i-2}+x_{2i}}{3}), & x_{2i-1} \leq x \leq x_{2i}. \end{cases}$$

(i) Calculate and simplify $p^{(n)}(x)$ for $n = 1,2,3,4$. Use the simplified expression for $p'(x)$ to calculate $p''(x)$, and so on.

(ii) Verify that $p(x_{2i-2}) = p'(x_{2i-2}) = p''(x_{2i-2}) = p(x_{2i}) = p'(x_{2i}) = p''(x_{2i}) = 0$.

[1] This function is constructed to have the properties listed below, so that certain terms vanish upon the repeated integration by parts we will perform later. We could construct this function from scratch, but it is a lot of algebra and would be very time-consuming.

(iii) Show that $p^{(n)}(x)$ for $n = 0, 1, 2$ is continuous on $[x_{2i-2}, x_{2i}]$. (*Hint*: $x_{2i-1} = \frac{x_{2i-2}+x_{2i}}{2}$.)

(iv) Show that $p^{(3)}(x)$ is continuous on $[x_{2i-2}, x_{2i-1}]$ and $[x_{2i-1}, x_{2i}]$, but has a jump discontinuity at x_{2i-1}.

(v) Calculate $p^{(3)}(x_{2i-2})$ and $p^{(3)}(x_{2i})$.

(b) Consider now the integral $\int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx$. Integrating by parts repeatedly,

$$\int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx = f^{(3)}(x)p(x)\Big|_{x_{2i-2}}^{x_{2i}} - f^{(2)}(x)p'(x)\Big|_{x_{2i-2}}^{x_{2i}} + f'(x)p''(x)\Big|_{x_{2i-2}}^{x_{2i}} - \int_{x_{2i-2}}^{x_{2i}} f'(x)p^{(3)}(x)dx.$$

Since $f(x)p^{(3)}(x)$ is continuous except for a single jump discontinuity, it is integrable. In order to use the Fundamental Theorem of Calculus to calculate the integral, we write it as

$$\int_{x_{2i-2}}^{x_{2i}} f'(x)p^{(3)}(x)dx = \int_{x_{2i-2}}^{x_{2i-1}} f'(x)p^{(3)}(x)dx + \int_{x_{2i-1}}^{x_{2i}} f'(x)p^{(3)}(x)dx,$$

so that each integrand is continuous over the corresponding subinterval. Therefore

$$\begin{aligned}\int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx = {} & f^{(3)}(x)p(x)\Big|_{x_{2i-2}}^{x_{2i}} - f^{(2)}(x)p'(x)\Big|_{x_{2i-2}}^{x_{2i}} + f'(x)p''(x)\Big|_{x_{2i-2}}^{x_{2i}} \\ & - \int_{x_{2i-2}}^{x_{2i-1}} f'(x)p^{(3)}(x)dx - \int_{x_{2i-1}}^{x_{2i}} f'(x)p^{(3)}(x)dx.\end{aligned}$$

Integrating each of the two remaining integrals by parts once more gives

$$\begin{aligned}\int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx = {} & f^{(3)}(x)p(x)\Big|_{x_{2i-2}}^{x_{2i}} - f^{(2)}(x)p'(x)\Big|_{x_{2i-2}}^{x_{2i}} + f'(x)p''(x)\Big|_{x_{2i-2}}^{x_{2i}} \\ & - f(x)p^{(3)}(x)\Big|_{x_{2i-2}}^{x_{2i-1}} - f(x)p^{(3)}(x)\Big|_{x_{2i-1}}^{x_{2i}} + \int_{x_{2i-2}}^{x_{2i-1}} f(x)p^{(4)}(x)dx \\ & + \int_{x_{2i-1}}^{x_{2i}} f(x)p^{(4)}(x)dx.\end{aligned}$$

By Problem 7(f), $f(x)p^{(4)}(x)$ is defined and is continuous at all points of $[x_{2i-2}, x_{2i}]$ except for x_{2i-1}, where it is undefined. Since changing the value of a continuous function at a single point does not change the integral, we have

$$\int_{x_{2i-2}}^{x_{2i-1}} f(x)p^{(4)}(x)dx + \int_{x_{2i-1}}^{x_{2i}} f(x)p^{(4)}(x)dx = \int_{x_{2i-2}}^{x_{2i}} f(x)p^{(4)}(x)dx$$

and therefore

$$\begin{aligned}\int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx = {} & f^{(3)}(x)p(x)\Big|_{x_{2i-2}}^{x_{2i}} - f^{(2)}(x)p'(x)\Big|_{x_{2i-2}}^{x_{2i}} + f'(x)p''(x)\Big|_{x_{2i-2}}^{x_{2i}} \\ & - f(x)p^{(3)}(x)\Big|_{x_{2i-2}}^{x_{2i-1}} - f(x)p^{(3)}(x)\Big|_{x_{2i-1}}^{x_{2i}} + \int_{x_{2i-2}}^{x_{2i}} f(x)p^{(4)}(x)dx \\ = {} & f^{(3)}(x)p(x)\Big|_{x_{2i-2}}^{x_{2i}} - f^{(2)}(x)p'(x)\Big|_{x_{2i-2}}^{x_{2i}} + f'(x)p''(x)\Big|_{x_{2i-2}}^{x_{2i}} \\ & - f(x)p^{(3)}(x)\Big|_{x_{2i-2}}^{x_{2i-1}} - f(x)p^{(3)}(x)\Big|_{x_{2i-1}}^{x_{2i}} + 24\int_{x_{2i-2}}^{x_{2i}} f(x)dx.\end{aligned}$$

(c) Use your answers from Problem 7(b)-(f) to show that this implies

$$24\int_{x_{2i-2}}^{x_{2i}} f(x)dx = -f(x)p^{(3)}(x)\Big|_{x_{2i-2}}^{x_{2i-1}} - f(x)p^{(3)}(x)\Big|_{x_{2i-1}}^{x_{2i}} + \int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx$$

and therefore

$$24\int_{x_{2i-2}}^{x_{2i}} f(x)dx = -f(x_{2i-1})p^{(3)}(x_{2i-1}^{-}) + f(x_{2i-2})p^{(3)}(x_{2i-2}) - f(x_{2i})p^{(3)}(x_{2i}) + f(x_{2i-1})p^{(3)}(x_{2i-1}^{+}) \int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx.$$

(d) Again, use your answers from 7(b)-(f) to show that this expression simplifies to

$$\int_{x_{2i-2}}^{x_{2i}} f(x)dx = \frac{x_{2i}-x_{2i-2}}{6}\left(f(x_{2i-2}) + 4f(x_{2i-1}) + f(x_{2i})\right) + \frac{1}{24}\int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx.$$

It follows that

$$\int_a^b f(x)dx = \sum_{i=1}^{n/2} \int_{x_{2i-2}}^{x_{2i}} f(x)dx$$
$$= \sum_{i=1}^{n/2} \frac{x_{2i} - x_{2i-2}}{6} \left(f(x_{2i-2}) + 4f(x_{2i-1}) + f(x_{2i})\right) + \frac{1}{24} \sum_{i=1}^{n/2} \int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx.$$

(e) Show that the expression above implies that $|E_n^S(f)| \leq \frac{M}{24} \sum_{i=1}^{n/2} \int_{x_{2i-2}}^{x_{2i}} |p(x)|dx$.

Using the definition of $p(x)$, we can write

$$\begin{aligned}
&\frac{M}{24}\sum_{i=1}^{n/2}\int_{x_{2i-2}}^{x_{2i}}|p(x)|dx \\
&=\frac{M}{24}\sum_{i=1}^{n/2}\left[\int_{x_{2i-2}}^{x_{2i-1}}\left|(x-x_{2i-2})^3(x-\frac{x_{2i-2}}{3}-\frac{2x_{2i}}{3})\right|dx+\int_{x_{2i-1}}^{x_{2i}}\left|(x-x_{2i})^3(x-\frac{2x_{2i-2}}{3}-\frac{x_{2i}}{3})\right|dx\right] \\
&=\frac{M}{24}\sum_{i=1}^{n/2}\left[\int_{x_{2i-2}}^{x_{2i-1}}|x-x_{2i-2}|^3\left|x-\frac{x_{2i-2}}{3}-\frac{2x_{2i}}{3}\right|dx+\int_{x_{2i-1}}^{x_{2i}}|x-x_{2i}|^3\left|x-\frac{2x_{2i-2}}{3}-\frac{x_{2i}}{3}\right|dx\right] \\
&=\frac{M}{24}\sum_{i=1}^{n/2}\left[\int_{x_{2i-2}}^{x_{2i-1}}(x-x_{2i-2})^3\left|x-\frac{x_{2i-2}}{3}-\frac{2x_{2i}}{3}\right|dx+\int_{x_{2i-1}}^{x_{2i}}(x_{2i}-x)^3\left|x-\frac{2x_{2i-2}}{3}-\frac{x_{2i}}{3}\right|dx\right].
\end{aligned}$$

(a) Show that for $x_{2i-2}\le x\le x_{2i-1}$, $x-\frac{x_{2i-2}}{3}-\frac{2x_{2i}}{3}<0$, and therefore $\left|x-\frac{x_{2i-2}}{3}-\frac{2x_{2i}}{3}\right|=\frac{x_{2i-2}}{3}+\frac{2x_{2i}}{3}-x$. (*Hint*: Recall that $x_{2i-1}=\frac{x_{2i-2}+x_{2i}}{2}$.)

(b) Show that for $x_{2i-1}\le x\le x_{2i}$, $x-\frac{2x_{2i-2}}{3}-\frac{x_{2i}}{3}>0$ and therefore $\left|x-\frac{2x_{2i-2}}{3}-\frac{x_{2i}}{3}\right|=x-\frac{2x_{2i-2}}{3}-\frac{x_{2i}}{3}$.

It follows from the previous problem that

$$\int_{x_{2i-2}}^{x_{2i-1}} (x-x_{2i-2})^3 \left| x - \frac{x_{2i-2}}{3} - \frac{2x_{2i}}{3} \right| dx + \int_{x_{2i-1}}^{x_{2i}} (x_{2i}-x)^3 \left| x - \frac{2x_{2i-2}}{3} - \frac{x_{2i}}{3} \right| dx$$
$$= \int_{x_{2i-2}}^{x_{2i-1}} (x-x_{2i-2})^3 \left(\frac{x_{2i-2}}{3} + \frac{2x_{2i}}{3} - x \right) dx + \int_{x_{2i-1}}^{x_{2i}} (x_{2i}-x)^3 \left(x - \frac{2x_{2i-2}}{3} - \frac{x_{2i}}{3} \right) dx$$

(f) Show that

$$\int_{x_{2i-2}}^{x_{2i-1}} (x-x_{2i-2})^3 \left(\frac{x_{2i-2}}{3} + \frac{2x_{2i}}{3} - x \right) dx = \int_{x_{2i-1}}^{x_{2i}} (x_{2i}-x)^3 \left(x - \frac{2x_{2i-2}}{3} - \frac{x_{2i}}{3} \right) dx$$

and therefore

$$\int_{x_{2i-2}}^{x_{2i-1}} (x-x_{2i-2})^3 \left(\frac{x_{2i-2}}{3} + \frac{2x_{2i}}{3} - x \right) dx + \int_{x_{2i-1}}^{x_{2i}} (x_{2i}-x)^3 \left(x - \frac{2x_{2i-2}}{3} - \frac{x_{2i}}{3} \right) dx$$
$$= 2\int_{x_{2i-2}}^{x_{2i-1}} (x-x_{2i-2})^3 \left(\frac{x_{2i-2}}{3} + \frac{2x_{2i}}{3} - x \right) dx$$

by making the substitution $u = x_{2i-2} + x_{2i} - x$ in $\int_{x_{2i-1}}^{x_{2i}} (x_{2i}-x)^3 \left(x - \frac{2x_{2i-2}}{3} - \frac{x_{2i}}{3} \right) dx$.

(g) Show that $\frac{x_{2i-2}}{3}+\frac{2x_{2i}}{3}-x=\frac{2}{3}(x_{2i}-x_{2i-2})-(x-x_{2i-2})$ and therefore

$$\int_{x_{2i-2}}^{x_{2i-1}}(x-x_{2i-2})^3\left(\frac{x_{2i-2}}{3}+\frac{2x_{2i}}{3}-x\right)dx=\int_{x_{2i-2}}^{x_{2i-1}}(x-x_{2i-2})^3\left(\frac{2}{3}(x_{2i}-x_{2i-2})-(x-x_{2i-2})\right)dx.$$

(*Hint*: $\frac{x_{2i-2}}{3}+\frac{2x_{2i}}{3}-x=\frac{x_{2i-2}}{3}+\frac{2x_{2i}}{3}-x+x_{2i-2}-\frac{3x_{2i-2}}{3}$)

(h) Show that $\int_{x_{2i-2}}^{x_{2i-1}}(x-x_{2i-2})^3\left(\frac{2}{3}(x_{2i}-x_{2i-2})-(x-x_{2i-2})\right)dx=\frac{1}{240}(x_{2i}-x_{2i-2})^5$.

(i) Put everything together to complete the proof.

□